

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

Device Specifications

Device Type: Audio Processor
Manufacturer: Extron
Firmware Version: N/A
Model(s): SSP 200

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scriptor Version
3.12.0000-b001	2.8.0

Version History

Module Version	Date	Notes
1_0_0_0	12/31/2020	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
Example: `InterfaceName.connectionCounter = 5`
- If login credentials are required, devicePassword and deviceUsername must be set accordingly. deviceUsername default value is admin respectively.
Example: `InterfaceName.deviceUsername = 'extron'`

Supported Classes and Examples

SerialClass

```
InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='SSP 200')
```

SerialOverEthernetClass

```
InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='SSP 200')
```

SSHClass

#Password Required

```
InterfaceName = ModuleName.SSHClass('192.168.254.254', 22023, Credentials=('admin', 'extron'),  
Model='SSP 200')
```

#No Password Required

```
InterfaceName = ModuleName.SSHClass('192.168.254.254', 22023, Credentials=('admin', ''),  
Model='SSP 200')
```

Control Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command	Value		
AnalogInputGain	-18 to 24 in steps of 1		
# AnalogInputGain example InterfaceName.Set('AnalogInputGain', 24)			
Command	Value		Value
AudioMute	'On'	'Off'	
Qualifier Key	Qualifier Value		
'Output'	'1' – '16'		
# AudioMute example InterfaceName.Set('AudioMute', 'On', {'Output': '1'})			
Command	Value	Value	Value
ExecutiveMode	'Mode 1'	'Mode 2'	'Mode 3'
	'Off'		
# ExecutiveMode example InterfaceName.Set('ExecutiveMode', 'Mode 1')			
Command	Value	Value	
GlobalAudioMute	'On'	'Off'	
# GlobalAudioMute example InterfaceName.Set('GlobalAudioMute', 'On')			
Command	Value		
Input	'1' – '5'		
# Input example InterfaceName.Set('Input', '1')			
Command	Value	Value	Value
ListeningMode	'Auto'	'Stereo'	'Mono'
	'Stereo to all'	'Mono to all'	'Dolby Surround'
	'DTS Neural:X'		
Qualifier Key	Qualifier Value		
'Input'	'1' – '5'		
Qualifier Key	Qualifier Value	Qualifier Value	Qualifier Value
'Source Format'	'Not Present'	'Analog'	'PCM'
	'Dolby Digital'	'Dolby Digital Plus'	'Dolby Digital Plus ATMOS'
	'Dolby TrueHD'	'Dolby TrueHD ATMOS'	'DTS or DTS 96/24'
	'DTS-ES (96/24, Matrix, or Discrete)'	'DTS-HD High Resolution Audio'	'DTS-HD Master Audio'
	'DTS Express'	'DTS:X'	
# ListeningMode example InterfaceName.Set('ListeningMode', 'Auto', {'Input': '1', 'Source Format': 'Not Present'})			
Command	Value		
OutputTrim	-24 to 0 in steps of 1		
Qualifier Key	Qualifier Value		
'Output'	'1' – '16'		
# OutputTrim example InterfaceName.Set('OutputTrim', 0, {'Output': '1'})			

**Global Scripter Module
Communication Sheet**

Command Subwoofer	Value 'On'	Value 'Off'
# Subwoofer example InterfaceName.Set('Subwoofer', 'On')		
Command Volume	Value 0 to 100 in steps of 1	
# Volume example InterfaceName.Set('Volume', 100)		

Status Available

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'},
```

FeedbackHandler)

FeedbackHandler will be called only when the specified qualifier gets a new status.

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command	Value		
AnalogInputGain	-18 to 24 in steps of 1		
# AnalogInputGain example InterfaceName.Update('AnalogInputGain') Value = InterfaceName.ReadStatus('AnalogInputGain') InterfaceName.SubscribeStatus('AnalogInputGain', None, FeedbackHandler)			
Command	Value	Value	
AudioMute	'On'	'Off'	
Qualifier Key	Qualifier Value		
'Output'	'1' – '16'		
# AudioMute example InterfaceName.Update('AudioMute', {'Output': '1'}) Value = InterfaceName.ReadStatus('AudioMute', {'Output': '1'}) InterfaceName.SubscribeStatus('AudioMute', None, FeedbackHandler)			
Command	Value	Value	
ConnectionStatus	'Connected'	'Disconnected'	
# ConnectionStatus example Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)			
Command	Value	Value	Value
ExecutiveMode	'Mode 1'	'Mode 2'	'Mode 3'
	'Off'		
# ExecutiveMode example InterfaceName.Update('ExecutiveMode') Value = InterfaceName.ReadStatus('ExecutiveMode') InterfaceName.SubscribeStatus('ExecutiveMode', None, FeedbackHandler)			
Command	Value		
EXPIInputAssignment	'0' – '4'		
# EXPIInputAssignment example InterfaceName.Update('EXPIInputAssignment') Value = InterfaceName.ReadStatus('EXPIInputAssignment') InterfaceName.SubscribeStatus('EXPIInputAssignment', None, FeedbackHandler)			
Command	Value		
Input	'1' – '5'		
# Input example InterfaceName.Update('Input') Value = InterfaceName.ReadStatus('Input') InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)			
Command	Value	Value	Value

Global Scriptor Module Communication Sheet

ListeningMode	'Auto' 'Stereo to all' 'DTS Neural:X'	'Stereo' 'Mono to all'	'Mono' 'Dolby Surround'
Qualifier Key 'Input'	Qualifier Value '1' – '5'		
Qualifier Key 'Source Format'	Qualifier Value 'Not Present' 'Dolby Digital' 'Dolby TrueHD' 'DTS-ES (96/24, Matrix, or Discrete)' 'DTS Express'	Qualifier Value 'Analog' 'Dolby Digital Plus' 'Dolby TrueHD ATMOS' 'DTS-HD High Resolution Audio' 'DTS:X'	Qualifier Value 'PCM' 'Dolby Digital Plus ATMOS' 'DTS or DTS 96/24' 'DTS-HD Master Audio'
# ListeningMode example InterfaceName.Update('ListeningMode', {'Input': '1', 'Source Format': 'Not Present'}) Value = InterfaceName.ReadStatus('ListeningMode', {'Input': '1', 'Source Format': 'Not Present'}) InterfaceName.SubscribeStatus('ListeningMode', None, FeedbackHandler)			
Command OutputTrim	Value -24 to 0 in steps of 1		
Qualifier Key 'Output'	Qualifier Value '1' – '16'		
# OutputTrim example InterfaceName.Update('OutputTrim', {'Output': '1'}) Value = InterfaceName.ReadStatus('OutputTrim', {'Output': '1'}) InterfaceName.SubscribeStatus('OutputTrim', None, FeedbackHandler)			
Command SourceFormat	Value 'Not Present' 'Dolby Digital' 'Dolby TrueHD' 'DTS-ES (96/24, Matrix, or Discrete)' 'DTS Express'	Value 'Analog' 'Dolby Digital Plus' 'Dolby TrueHD ATMOS' 'DTS-HD High Resolution Audio' 'DTS:X'	Value 'PCM' 'Dolby Digital Plus ATMOS' 'DTS or DTS 96/24' 'DTS-HD Master Audio'
# SourceFormat example InterfaceName.Update('SourceFormat') Value = InterfaceName.ReadStatus('SourceFormat') InterfaceName.SubscribeStatus('SourceFormat', None, FeedbackHandler)			
Command Subwoofer	Value 'On'	Value 'Off'	
# Subwoofer example InterfaceName.Update('Subwoofer') Value = InterfaceName.ReadStatus('Subwoofer') InterfaceName.SubscribeStatus('Subwoofer', None, FeedbackHandler)			
Command Volume	Value 0 to 100 in steps of 1		
# Volume example InterfaceName.Update('Volume') Value = InterfaceName.ReadStatus('Volume') InterfaceName.SubscribeStatus('Volume', None, FeedbackHandler)			

Cable and Adapter Requirements

Captive Screw to Captive Screw

Notes for the Device

Serial communication

Port Type: RS-232

Baud Rate: 9600

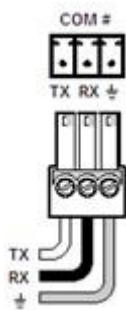
Data Bits: 8

Parity: None

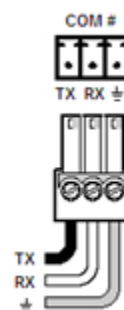
Stop Bits: One

Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Signal
TxD		TxD
RxD		RxD
GND		GND



Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scriptor ethernet interface.

Port Type: Ethernet

Default Port: 22023

Logon Credentials Supported: Yes

Default Username: admin

Multi-Connection Yes

Capabilities:

Port Changeability: Yes

Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

Notes for the Device

Appendix A. Set Commands

Analog Input Gain -18	\x1Ba-18SSP\x0D
Analog Input Gain 24	\x1Ba24SSP\x0D
Audio Mute Off Output 1	1*0Z
Audio Mute Off Output 12	12*0Z
Audio Mute Off Output 15	15*0Z
Audio Mute Off Output 16	16*0Z
Audio Mute On Output 1	1*1Z
Audio Mute On Output 12	12*1Z
Audio Mute On Output 15	15*1Z
Audio Mute On Output 16	16*1Z
Executive Mode Mode 1	1X
Executive Mode Mode 2	2X
Executive Mode Mode 3	3X
Executive Mode Off	0X
Global Audio Mute Off	0Z
Global Audio Mute On	1Z
Input 1	1\$
Input 5	5\$
Listening Mode Auto Input 1 Source Format Analog	\x1B11*1*1SSP\x0D
Listening Mode Auto Input 1 Source Format Dolby Digital	\x1B11*3*1SSP\x0D
Listening Mode Auto Input 1 Source Format Dolby Digital Plus	\x1B11*4*1SSP\x0D
Listening Mode Auto Input 1 Source Format Dolby Digital Plus ATMOS	\x1B11*5*1SSP\x0D
Listening Mode Auto Input 1 Source Format Dolby TrueHD	\x1B11*6*1SSP\x0D
Listening Mode Auto Input 1 Source Format Dolby TrueHD ATMOS	\x1B11*7*1SSP\x0D
Listening Mode Auto Input 1 Source Format DTS Express	\x1B11*12*1SSP\x0D
Listening Mode Auto Input 1 Source Format DTS or DTS 96/24	\x1B11*8*1SSP\x0D
Listening Mode Auto Input 1 Source Format DTS:X	\x1B11*13*1SSP\x0D
Listening Mode Auto Input 1 Source Format DTS-ES (96/24, Matrix, or Discrete)	\x1B11*9*1SSP\x0D
Listening Mode Auto Input 1 Source Format DTS-HD High Resolution Audio	\x1B11*10*1SSP\x0D
Listening Mode Auto Input 1 Source Format DTS-HD Master Audio	\x1B11*11*1SSP\x0D
Listening Mode Auto Input 1 Source Format Not Present	\x1B11*0*1SSP\x0D
Listening Mode Auto Input 1 Source Format PCM	\x1B11*2*1SSP\x0D
Listening Mode Auto Input 5 Source Format Analog	\x1B15*1*1SSP\x0D

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Listening Mode Auto Input 5 Source Format Dolby Digital	\x1B15*3*1SSP\x0D
Listening Mode Auto Input 5 Source Format Dolby Digital Plus	\x1B15*4*1SSP\x0D
Listening Mode Auto Input 5 Source Format Dolby Digital Plus ATMOS	\x1B15*5*1SSP\x0D
Listening Mode Auto Input 5 Source Format Dolby TrueHD	\x1B15*6*1SSP\x0D
Listening Mode Auto Input 5 Source Format Dolby TrueHD ATMOS	\x1B15*7*1SSP\x0D
Listening Mode Auto Input 5 Source Format DTS Express	\x1B15*12*1SSP\x0D
Listening Mode Auto Input 5 Source Format DTS or DTS 96/24	\x1B15*8*1SSP\x0D
Listening Mode Auto Input 5 Source Format DTS:X	\x1B15*13*1SSP\x0D
Listening Mode Auto Input 5 Source Format DTS-ES (96/24, Matrix, or Discrete)	\x1B15*9*1SSP\x0D
Listening Mode Auto Input 5 Source Format DTS-HD High Resolution Audio	\x1B15*10*1SSP\x0D
Listening Mode Auto Input 5 Source Format DTS-HD Master Audio	\x1B15*11*1SSP\x0D
Listening Mode Auto Input 5 Source Format Not Present	\x1B15*0*1SSP\x0D
Listening Mode Auto Input 5 Source Format PCM	\x1B15*2*1SSP\x0D
Listening Mode Dolby Surround Input 1 Source Format Analog	\x1B11*1*6SSP\x0D
Listening Mode Dolby Surround Input 1 Source Format Dolby Digital	\x1B11*3*6SSP\x0D
Listening Mode Dolby Surround Input 1 Source Format Dolby Digital Plus	\x1B11*4*6SSP\x0D
Listening Mode Dolby Surround Input 1 Source Format Dolby Digital Plus ATMOS	\x1B11*5*6SSP\x0D
Listening Mode Dolby Surround Input 1 Source Format Dolby TrueHD	\x1B11*6*6SSP\x0D
Listening Mode Dolby Surround Input 1 Source Format Dolby TrueHD ATMOS	\x1B11*7*6SSP\x0D
Listening Mode Dolby Surround Input 1 Source Format DTS Express	\x1B11*12*6SSP\x0D
Listening Mode Dolby Surround Input 1 Source Format DTS or DTS 96/24	\x1B11*8*6SSP\x0D
Listening Mode Dolby Surround Input 1 Source Format DTS:X	\x1B11*13*6SSP\x0D
Listening Mode Dolby Surround Input 1 Source Format DTS-ES (96/24, Matrix, or Discrete)	\x1B11*9*6SSP\x0D
Listening Mode Dolby Surround Input 1 Source Format DTS-HD High Resolution Audio	\x1B11*10*6SSP\x0D
Listening Mode Dolby Surround Input 1 Source	\x1B11*11*6SSP\x0D

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Format DTS-HD Master Audio	
Listening Mode Dolby Surround Input 1 Source Format Not Present	\x1B11*0*6SSP\x0D
Listening Mode Dolby Surround Input 1 Source Format PCM	\x1B11*2*6SSP\x0D
Listening Mode Dolby Surround Input 5 Source Format Analog	\x1B15*1*6SSP\x0D
Listening Mode Dolby Surround Input 5 Source Format Dolby Digital	\x1B15*3*6SSP\x0D
Listening Mode Dolby Surround Input 5 Source Format Dolby Digital Plus	\x1B15*4*6SSP\x0D
Listening Mode Dolby Surround Input 5 Source Format Dolby Digital Plus ATMOS	\x1B15*5*6SSP\x0D
Listening Mode Dolby Surround Input 5 Source Format Dolby TrueHD	\x1B15*6*6SSP\x0D
Listening Mode Dolby Surround Input 5 Source Format Dolby TrueHD ATMOS	\x1B15*7*6SSP\x0D
Listening Mode Dolby Surround Input 5 Source Format DTS Express	\x1B15*12*6SSP\x0D
Listening Mode Dolby Surround Input 5 Source Format DTS or DTS 96/24	\x1B15*8*6SSP\x0D
Listening Mode Dolby Surround Input 5 Source Format DTS:X	\x1B15*13*6SSP\x0D
Listening Mode Dolby Surround Input 5 Source Format DTS-ES (96/24, Matrix, or Discrete)	\x1B15*9*6SSP\x0D
Listening Mode Dolby Surround Input 5 Source Format DTS-HD High Resolution Audio	\x1B15*10*6SSP\x0D
Listening Mode Dolby Surround Input 5 Source Format DTS-HD Master Audio	\x1B15*11*6SSP\x0D
Listening Mode Dolby Surround Input 5 Source Format Not Present	\x1B15*0*6SSP\x0D
Listening Mode Dolby Surround Input 5 Source Format PCM	\x1B15*2*6SSP\x0D
Listening Mode DTS Neural:X Input 1 Source Format Analog	\x1B11*1*7SSP\x0D
Listening Mode DTS Neural:X Input 1 Source Format Dolby Digital	\x1B11*3*7SSP\x0D
Listening Mode DTS Neural:X Input 1 Source Format Dolby Digital Plus	\x1B11*4*7SSP\x0D
Listening Mode DTS Neural:X Input 1 Source Format Dolby Digital Plus ATMOS	\x1B11*5*7SSP\x0D
Listening Mode DTS Neural:X Input 1 Source Format Dolby TrueHD	\x1B11*6*7SSP\x0D
Listening Mode DTS Neural:X Input 1 Source Format Dolby TrueHD ATMOS	\x1B11*7*7SSP\x0D
Listening Mode DTS Neural:X Input 1 Source Format DTS Express	\x1B11*12*7SSP\x0D

Global Scripter Module Communication Sheet

Listening Mode DTS Neural:X Input 1 Source Format DTS or DTS 96/24	\x1B11*8*7SSP\x0D
Listening Mode DTS Neural:X Input 1 Source Format DTS:X	\x1B11*13*7SSP\x0D
Listening Mode DTS Neural:X Input 1 Source Format DTS-ES (96/24, Matrix, or Discrete)	\x1B11*9*7SSP\x0D
Listening Mode DTS Neural:X Input 1 Source Format DTS-HD High Resolution Audio	\x1B11*10*7SSP\x0D
Listening Mode DTS Neural:X Input 1 Source Format DTS-HD Master Audio	\x1B11*11*7SSP\x0D
Listening Mode DTS Neural:X Input 1 Source Format Not Present	\x1B11*0*7SSP\x0D
Listening Mode DTS Neural:X Input 1 Source Format PCM	\x1B11*2*7SSP\x0D
Listening Mode DTS Neural:X Input 5 Source Format Analog	\x1B15*1*7SSP\x0D
Listening Mode DTS Neural:X Input 5 Source Format Dolby Digital	\x1B15*3*7SSP\x0D
Listening Mode DTS Neural:X Input 5 Source Format Dolby Digital Plus	\x1B15*4*7SSP\x0D
Listening Mode DTS Neural:X Input 5 Source Format Dolby Digital Plus ATMOS	\x1B15*5*7SSP\x0D
Listening Mode DTS Neural:X Input 5 Source Format Dolby TrueHD	\x1B15*6*7SSP\x0D
Listening Mode DTS Neural:X Input 5 Source Format Dolby TrueHD ATMOS	\x1B15*7*7SSP\x0D
Listening Mode DTS Neural:X Input 5 Source Format DTS Express	\x1B15*12*7SSP\x0D
Listening Mode DTS Neural:X Input 5 Source Format DTS or DTS 96/24	\x1B15*8*7SSP\x0D
Listening Mode DTS Neural:X Input 5 Source Format DTS:X	\x1B15*13*7SSP\x0D
Listening Mode DTS Neural:X Input 5 Source Format DTS-ES (96/24, Matrix, or Discrete)	\x1B15*9*7SSP\x0D
Listening Mode DTS Neural:X Input 5 Source Format DTS-HD High Resolution Audio	\x1B15*10*7SSP\x0D
Listening Mode DTS Neural:X Input 5 Source Format DTS-HD Master Audio	\x1B15*11*7SSP\x0D
Listening Mode DTS Neural:X Input 5 Source Format Not Present	\x1B15*0*7SSP\x0D
Listening Mode DTS Neural:X Input 5 Source Format PCM	\x1B15*2*7SSP\x0D
Listening Mode Mono Input 1 Source Format Analog	\x1B11*1*3SSP\x0D
Listening Mode Mono Input 1 Source Format Dolby Digital	\x1B11*3*3SSP\x0D
Listening Mode Mono Input 1 Source Format Dolby Digital Plus	\x1B11*4*3SSP\x0D

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Listening Mode Mono Input 1 Source Format Dolby Digital Plus ATMOS	\x1B11*5*3SSP\x0D
Listening Mode Mono Input 1 Source Format Dolby TrueHD	\x1B11*6*3SSP\x0D
Listening Mode Mono Input 1 Source Format Dolby TrueHD ATMOS	\x1B11*7*3SSP\x0D
Listening Mode Mono Input 1 Source Format DTS Express	\x1B11*12*3SSP\x0D
Listening Mode Mono Input 1 Source Format DTS or DTS 96/24	\x1B11*8*3SSP\x0D
Listening Mode Mono Input 1 Source Format DTS:X	\x1B11*13*3SSP\x0D
Listening Mode Mono Input 1 Source Format DTS-ES (96/24, Matrix, or Discrete)	\x1B11*9*3SSP\x0D
Listening Mode Mono Input 1 Source Format DTS-HD High Resolution Audio	\x1B11*10*3SSP\x0D
Listening Mode Mono Input 1 Source Format DTS-HD Master Audio	\x1B11*11*3SSP\x0D
Listening Mode Mono Input 1 Source Format Not Present	\x1B11*0*3SSP\x0D
Listening Mode Mono Input 1 Source Format PCM	\x1B11*2*3SSP\x0D
Listening Mode Mono Input 5 Source Format Analog	\x1B15*1*3SSP\x0D
Listening Mode Mono Input 5 Source Format Dolby Digital	\x1B15*3*3SSP\x0D
Listening Mode Mono Input 5 Source Format Dolby Digital Plus	\x1B15*4*3SSP\x0D
Listening Mode Mono Input 5 Source Format Dolby Digital Plus ATMOS	\x1B15*5*3SSP\x0D
Listening Mode Mono Input 5 Source Format Dolby TrueHD	\x1B15*6*3SSP\x0D
Listening Mode Mono Input 5 Source Format Dolby TrueHD ATMOS	\x1B15*7*3SSP\x0D
Listening Mode Mono Input 5 Source Format DTS Express	\x1B15*12*3SSP\x0D
Listening Mode Mono Input 5 Source Format DTS or DTS 96/24	\x1B15*8*3SSP\x0D
Listening Mode Mono Input 5 Source Format DTS:X	\x1B15*13*3SSP\x0D
Listening Mode Mono Input 5 Source Format DTS-ES (96/24, Matrix, or Discrete)	\x1B15*9*3SSP\x0D
Listening Mode Mono Input 5 Source Format DTS-HD High Resolution Audio	\x1B15*10*3SSP\x0D
Listening Mode Mono Input 5 Source Format DTS-HD Master Audio	\x1B15*11*3SSP\x0D
Listening Mode Mono Input 5 Source Format Not Present	\x1B15*0*3SSP\x0D
Listening Mode Mono Input 5 Source Format PCM	\x1B15*2*3SSP\x0D
Listening Mode Mono to all Input 1 Source Format Analog	\x1B11*1*5SSP\x0D

Global Scripter Module Communication Sheet

Listening Mode Mono to all Input 1 Source Format Dolby Digital	\x1B11*3*5SSP\x0D
Listening Mode Mono to all Input 1 Source Format Dolby Digital Plus	\x1B11*4*5SSP\x0D
Listening Mode Mono to all Input 1 Source Format Dolby Digital Plus ATMOS	\x1B11*5*5SSP\x0D
Listening Mode Mono to all Input 1 Source Format Dolby TrueHD	\x1B11*6*5SSP\x0D
Listening Mode Mono to all Input 1 Source Format Dolby TrueHD ATMOS	\x1B11*7*5SSP\x0D
Listening Mode Mono to all Input 1 Source Format DTS Express	\x1B11*12*5SSP\x0D
Listening Mode Mono to all Input 1 Source Format DTS or DTS 96/24	\x1B11*8*5SSP\x0D
Listening Mode Mono to all Input 1 Source Format DTS:X	\x1B11*13*5SSP\x0D
Listening Mode Mono to all Input 1 Source Format DTS-ES (96/24, Matrix, or Discrete)	\x1B11*9*5SSP\x0D
Listening Mode Mono to all Input 1 Source Format DTS-HD High Resolution Audio	\x1B11*10*5SSP\x0D
Listening Mode Mono to all Input 1 Source Format DTS-HD Master Audio	\x1B11*11*5SSP\x0D
Listening Mode Mono to all Input 1 Source Format Not Present	\x1B11*0*5SSP\x0D
Listening Mode Mono to all Input 1 Source Format PCM	\x1B11*2*5SSP\x0D
Listening Mode Mono to all Input 5 Source Format Analog	\x1B15*1*5SSP\x0D
Listening Mode Mono to all Input 5 Source Format Dolby Digital	\x1B15*3*5SSP\x0D
Listening Mode Mono to all Input 5 Source Format Dolby Digital Plus	\x1B15*4*5SSP\x0D
Listening Mode Mono to all Input 5 Source Format Dolby Digital Plus ATMOS	\x1B15*5*5SSP\x0D
Listening Mode Mono to all Input 5 Source Format Dolby TrueHD	\x1B15*6*5SSP\x0D
Listening Mode Mono to all Input 5 Source Format Dolby TrueHD ATMOS	\x1B15*7*5SSP\x0D
Listening Mode Mono to all Input 5 Source Format DTS Express	\x1B15*12*5SSP\x0D
Listening Mode Mono to all Input 5 Source Format DTS or DTS 96/24	\x1B15*8*5SSP\x0D
Listening Mode Mono to all Input 5 Source Format DTS:X	\x1B15*13*5SSP\x0D
Listening Mode Mono to all Input 5 Source Format DTS-ES (96/24, Matrix, or Discrete)	\x1B15*9*5SSP\x0D
Listening Mode Mono to all Input 5 Source Format	\x1B15*10*5SSP\x0D

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DTS-HD High Resolution Audio	
Listening Mode Mono to all Input 5 Source Format DTS-HD Master Audio	\x1B15*11*5SSP\x0D
Listening Mode Mono to all Input 5 Source Format Not Present	\x1B15*0*5SSP\x0D
Listening Mode Mono to all Input 5 Source Format PCM	\x1B15*2*5SSP\x0D
Listening Mode Stereo Input 1 Source Format Analog	\x1B11*1*2SSP\x0D
Listening Mode Stereo Input 1 Source Format Dolby Digital	\x1B11*3*2SSP\x0D
Listening Mode Stereo Input 1 Source Format Dolby Digital Plus	\x1B11*4*2SSP\x0D
Listening Mode Stereo Input 1 Source Format Dolby Digital Plus ATMOS	\x1B11*5*2SSP\x0D
Listening Mode Stereo Input 1 Source Format Dolby TrueHD	\x1B11*6*2SSP\x0D
Listening Mode Stereo Input 1 Source Format Dolby TrueHD ATMOS	\x1B11*7*2SSP\x0D
Listening Mode Stereo Input 1 Source Format DTS Express	\x1B11*12*2SSP\x0D
Listening Mode Stereo Input 1 Source Format DTS or DTS 96/24	\x1B11*8*2SSP\x0D
Listening Mode Stereo Input 1 Source Format DTS:X	\x1B11*13*2SSP\x0D
Listening Mode Stereo Input 1 Source Format DTS-ES (96/24, Matrix, or Discrete)	\x1B11*9*2SSP\x0D
Listening Mode Stereo Input 1 Source Format DTS-HD High Resolution Audio	\x1B11*10*2SSP\x0D
Listening Mode Stereo Input 1 Source Format DTS-HD Master Audio	\x1B11*11*2SSP\x0D
Listening Mode Stereo Input 1 Source Format Not Present	\x1B11*0*2SSP\x0D
Listening Mode Stereo Input 1 Source Format PCM	\x1B11*2*2SSP\x0D
Listening Mode Stereo Input 5 Source Format Analog	\x1B15*1*2SSP\x0D
Listening Mode Stereo Input 5 Source Format Dolby Digital	\x1B15*3*2SSP\x0D
Listening Mode Stereo Input 5 Source Format Dolby Digital Plus	\x1B15*4*2SSP\x0D
Listening Mode Stereo Input 5 Source Format Dolby Digital Plus ATMOS	\x1B15*5*2SSP\x0D
Listening Mode Stereo Input 5 Source Format Dolby TrueHD	\x1B15*6*2SSP\x0D
Listening Mode Stereo Input 5 Source Format Dolby TrueHD ATMOS	\x1B15*7*2SSP\x0D
Listening Mode Stereo Input 5 Source Format DTS Express	\x1B15*12*2SSP\x0D
Listening Mode Stereo Input 5 Source Format DTS or DTS 96/24	\x1B15*8*2SSP\x0D

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Listening Mode Stereo Input 5 Source Format DTS:X	\x1B15*13*2SSP\x0D
Listening Mode Stereo Input 5 Source Format DTS-ES (96/24, Matrix, or Discrete)	\x1B15*9*2SSP\x0D
Listening Mode Stereo Input 5 Source Format DTS-HD High Resolution Audio	\x1B15*10*2SSP\x0D
Listening Mode Stereo Input 5 Source Format DTS-HD Master Audio	\x1B15*11*2SSP\x0D
Listening Mode Stereo Input 5 Source Format Not Present	\x1B15*0*2SSP\x0D
Listening Mode Stereo Input 5 Source Format PCM	\x1B15*2*2SSP\x0D
Listening Mode Stereo to all Input 1 Source Format Analog	\x1B11*1*4SSP\x0D
Listening Mode Stereo to all Input 1 Source Format Dolby Digital	\x1B11*3*4SSP\x0D
Listening Mode Stereo to all Input 1 Source Format Dolby Digital Plus	\x1B11*4*4SSP\x0D
Listening Mode Stereo to all Input 1 Source Format Dolby Digital Plus ATMOS	\x1B11*5*4SSP\x0D
Listening Mode Stereo to all Input 1 Source Format Dolby TrueHD	\x1B11*6*4SSP\x0D
Listening Mode Stereo to all Input 1 Source Format Dolby TrueHD ATMOS	\x1B11*7*4SSP\x0D
Listening Mode Stereo to all Input 1 Source Format DTS Express	\x1B11*12*4SSP\x0D
Listening Mode Stereo to all Input 1 Source Format DTS or DTS 96/24	\x1B11*8*4SSP\x0D
Listening Mode Stereo to all Input 1 Source Format DTS:X	\x1B11*13*4SSP\x0D
Listening Mode Stereo to all Input 1 Source Format DTS-ES (96/24, Matrix, or Discrete)	\x1B11*9*4SSP\x0D
Listening Mode Stereo to all Input 1 Source Format DTS-HD High Resolution Audio	\x1B11*10*4SSP\x0D
Listening Mode Stereo to all Input 1 Source Format DTS-HD Master Audio	\x1B11*11*4SSP\x0D
Listening Mode Stereo to all Input 1 Source Format Not Present	\x1B11*0*4SSP\x0D
Listening Mode Stereo to all Input 1 Source Format PCM	\x1B11*2*4SSP\x0D
Listening Mode Stereo to all Input 5 Source Format Analog	\x1B15*1*4SSP\x0D
Listening Mode Stereo to all Input 5 Source Format Dolby Digital	\x1B15*3*4SSP\x0D
Listening Mode Stereo to all Input 5 Source Format Dolby Digital Plus	\x1B15*4*4SSP\x0D
Listening Mode Stereo to all Input 5 Source Format Dolby Digital Plus ATMOS	\x1B15*5*4SSP\x0D
Listening Mode Stereo to all Input 5 Source Format	\x1B15*6*4SSP\x0D

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Dolby TrueHD	
Listening Mode Stereo to all Input 5 Source Format Dolby TrueHD ATMOS	\x1B15*7*4SSP\x0D
Listening Mode Stereo to all Input 5 Source Format DTS Express	\x1B15*12*4SSP\x0D
Listening Mode Stereo to all Input 5 Source Format DTS or DTS 96/24	\x1B15*8*4SSP\x0D
Listening Mode Stereo to all Input 5 Source Format DTS:X	\x1B15*13*4SSP\x0D
Listening Mode Stereo to all Input 5 Source Format DTS-ES (96/24, Matrix, or Discrete)	\x1B15*9*4SSP\x0D
Listening Mode Stereo to all Input 5 Source Format DTS-HD High Resolution Audio	\x1B15*10*4SSP\x0D
Listening Mode Stereo to all Input 5 Source Format DTS-HD Master Audio	\x1B15*11*4SSP\x0D
Listening Mode Stereo to all Input 5 Source Format Not Present	\x1B15*0*4SSP\x0D
Listening Mode Stereo to all Input 5 Source Format PCM	\x1B15*2*4SSP\x0D
Output Trim 0 Output 1	\x1Bv1*0SSP\x0D
Output Trim 0 Output 12	\x1Bv12*0SSP\x0D
Output Trim 0 Output 15	\x1Bv15*0SSP\x0D
Output Trim 0 Output 16	\x1Bv16*0SSP\x0D
Output Trim -24 Output 1	\x1Bv1*-24SSP\x0D
Output Trim -24 Output 12	\x1Bv12*-24SSP\x0D
Output Trim -24 Output 15	\x1Bv15*-24SSP\x0D
Output Trim -24 Output 16	\x1Bv16*-24SSP\x0D
Subwoofer Off	\x1Bj0SSP\x0D
Subwoofer On	\x1Bj1SSP\x0D
Volume 0	0v
Volume 100	100v

Appendix B. Update Commands

Analog Input Gain	\x1BaSSP\x0D
Audio Mute Output 1	1*Z
Audio Mute Output 12	12*Z
Audio Mute Output 15	15*Z
Audio Mute Output 16	16*Z
Executive Mode	X
EXP Input Assignment	\x1BeSSP\x0D
Input	\$
Listening Mode Input 1 Source Format Analog	\x1B11*1SSP\x0D
Listening Mode Input 1 Source Format Dolby Digital	\x1B11*3SSP\x0D
Listening Mode Input 1 Source Format Dolby Digital Plus	\x1B11*4SSP\x0D
Listening Mode Input 1 Source Format Dolby Digital Plus ATMOS	\x1B11*5SSP\x0D
Listening Mode Input 1 Source Format Dolby TrueHD	\x1B11*6SSP\x0D
Listening Mode Input 1 Source Format Dolby TrueHD ATMOS	\x1B11*7SSP\x0D
Listening Mode Input 1 Source Format DTS Express	\x1B11*12SSP\x0D
Listening Mode Input 1 Source Format DTS or DTS 96/24	\x1B11*8SSP\x0D
Listening Mode Input 1 Source Format DTS:X	\x1B11*13SSP\x0D
Listening Mode Input 1 Source Format DTS-ES (96/24, Matrix, or Discrete)	\x1B11*9SSP\x0D
Listening Mode Input 1 Source Format DTS-HD High Resolution Audio	\x1B11*10SSP\x0D
Listening Mode Input 1 Source Format DTS-HD Master Audio	\x1B11*11SSP\x0D
Listening Mode Input 1 Source Format Not Present	\x1B11*0SSP\x0D
Listening Mode Input 1 Source Format PCM	\x1B11*2SSP\x0D
Listening Mode Input 5 Source Format Analog	\x1B15*1SSP\x0D
Listening Mode Input 5 Source Format Dolby Digital	\x1B15*3SSP\x0D
Listening Mode Input 5 Source Format Dolby Digital Plus	\x1B15*4SSP\x0D
Listening Mode Input 5 Source Format Dolby Digital Plus ATMOS	\x1B15*5SSP\x0D
Listening Mode Input 5 Source Format Dolby TrueHD	\x1B15*6SSP\x0D
Listening Mode Input 5 Source Format Dolby TrueHD ATMOS	\x1B15*7SSP\x0D
Listening Mode Input 5 Source Format DTS Express	\x1B15*12SSP\x0D
Listening Mode Input 5 Source Format DTS or DTS 96/24	\x1B15*8SSP\x0D
Listening Mode Input 5 Source Format DTS:X	\x1B15*13SSP\x0D
Listening Mode Input 5 Source Format DTS-ES (96/24, Matrix, or Discrete)	\x1B15*9SSP\x0D

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Listening Mode Input 5 Source Format DTS-HD High Resolution Audio	\x1B15*10SSP\x0D
Listening Mode Input 5 Source Format DTS-HD Master Audio	\x1B15*11SSP\x0D
Listening Mode Input 5 Source Format Not Present	\x1B15*0SSP\x0D
Listening Mode Input 5 Source Format PCM	\x1B15*2SSP\x0D
Output Trim Output 1	\x1Bv1SSP\x0D
Output Trim Output 12	\x1Bv12SSP\x0D
Output Trim Output 15	\x1Bv15SSP\x0D
Output Trim Output 16	\x1Bv16SSP\x0D
Source Format	33I
Subwoofer	\x1BjSSP\x0D
Volume	v